DO NOW · AO1 REVISION

Thermal conductivity and insulation

Energy
Paper 1 · Combined and Separate
7 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Describe how thermal conductivity affects the rate of energy transfer through a material.**	The higher the thermal conductivity, the higher the rate of energy transfer by conduction.	☐
2	**Name two things that affect how fast a building cools through its walls.**	The thickness and the thermal conductivity of the walls	☐
3	**Describe how the thickness of a building's walls affects its rate of energy loss.**	The thicker the walls, the lower the rate of energy loss.	☐
4	**Describe how the thermal conductivity of a building's walls affects its rate of energy loss.**	The higher the thermal conductivity, the higher the rate of energy loss.	☐
5	**Explain the purpose of using a good thermal insulator.**	To reduce unwanted energy transfers by heating.	☐
6 ★	**Explain why loft insulation reduces the heating bills of a house.**	Its low thermal conductivity reduces the rate of energy transfer out of the house.	☐
7 ★	**Give three things that affect the rate of energy transfer through a wall.**	Its thickness, its thermal conductivity and the temperature difference across it.	☐

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